



ET-SZ6 Automatic Voltage Regulator Operating Instructions

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1. Overview

ET-SZ6 on-load voltage regulation and control device (hereinafter referred to as ET-SZ6) is suitable for manual and automatic control of transformer's on-load voltage regulation, which supports at most 6 units to be used in parallel. ET-SZ6 has the functions of position and operation times display. It can also convert the position into telemetry signal 4-20mA and send it to the telemetry input port of RTU or through RS-485 (or RS-232) serial interface for remote transmission, as well as input through RS-485 (or RS-232) serial port to communicate with computer, controlling the on-load tap-changer to realize 1-N, N-1 and stop operations.

ET-SZ6 can also realize local manual operation of "1-N", "N-1" and "Stop", which is achieved by the buttons on the front panel.

2. Features

- a) The interface adopts large LCD screen
- b) Manual, automatic and remote control
- c) "1-N", "N-1" and "Stop" output relays are passive contacts
- d) Used as a single machine or at most 6 units in parallel
- e) Position display and operation times display
- f) Analog output 4 - 20mA
- g) Tap position signal BCD code output
- h) With RS-485 or RS-232 serial communication function
- i) The serial port communication protocol adopts Modbus-RTU protocol

3. Technical Parameters

- a) Auxiliary power supply: AC/DC 85-265V 50/60Hz
- b) Input mode: BCD input, decimal input
- c) Input range: 1 to 35 positions
- d) Control mode: automatic, manual and remote
- e) Parallel unit: 6 units
- f) Operation delay time: 1s to 600s adjustable, step size: 1s
- g) Signal voltage range: AC80 to 450V
- h) Sensitivity: 0.1% to 9.99% adjustable, step size: 0.01%
- i) Working temperature: - 20 °C to 50 °C
- j) Working humidity: less than 90% at 20 °C
- k) Overall dimension: 220×102×250mm
- l) Telemetry output: current: 4 - 20mA
- m) Remote signal output: BCD passive contact output (contact capacity AC220V/1A)
- n) Status output: passive contact 250V/1A

6. Connection

6.1 Diagram of the back panel

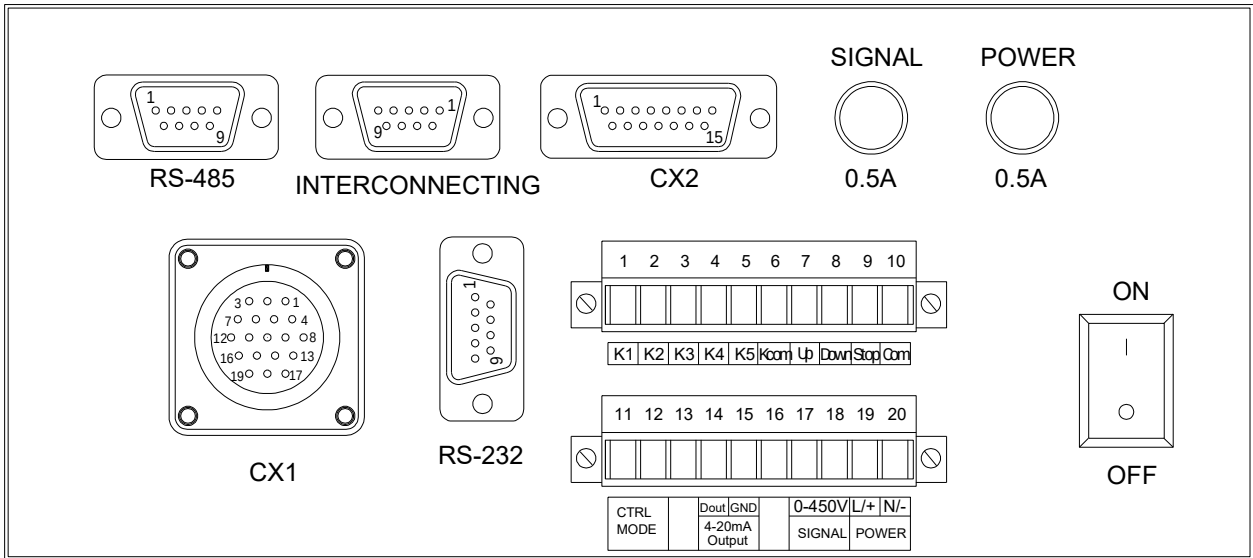


Fig. 2 Back panel – with sockets

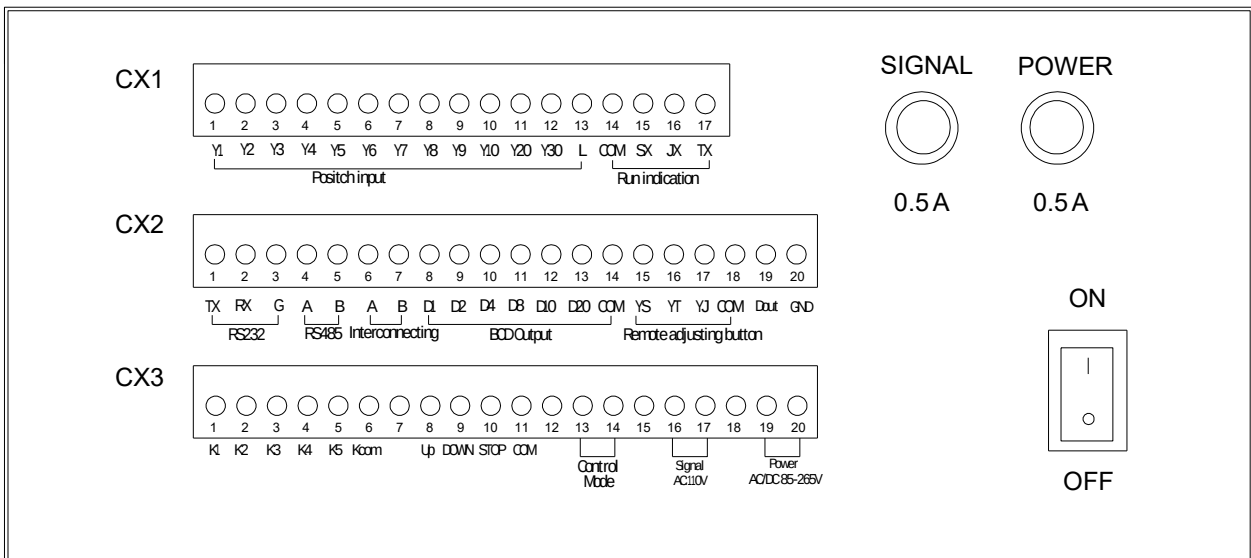


Fig. 3 Back panel – with terminals

D _{out}	The remote measuring output of position
GND	Analog output
Control mode	Passive contact point input. When breaking, it's manual control, and switches to manual control when closing. (The above input is valid only when the "control mode" is "external".)
Port	Used for exchanging the information when several units work in parallel.
Signal voltage	Connected to the signal voltage.
1-N, N-1, Stop, Common	Used for control output, and relay dead contact (AC250V/1A)

6.3 CX1 tap-changer position input connection table

No.	Socket type (decimal input)	Terminal type (BCD input)	Remark
CX - 1	Tap-changer position “1”	Tap-changer position “B01”	Connected to OLTC position signal input
CX - 2	Tap-changer position “2”	Tap-changer position “B02”	
CX - 3	Tap-changer position “3”	Empty	
CX - 4	Tap-changer position “4”	Tap-changer position “B04”	
CX - 5	Tap-changer position “5”	Empty	
CX - 6	Tap-changer position “6”	Empty	
CX - 7	Tap-changer position “7”	Empty	
CX - 8	Tap-changer position “8”	Tap-changer position “B08”	
CX - 9	Tap-changer position “9”	Empty	
CX - 10	Empty	Tap-changer position “B10”	
CX - 11	Empty	Tap-changer position “B20”	
CX - 12	Tap-changer position tens “1”		
CX - 13	Tap-changer position tens “2”	Position display common terminal L	
CX - 14	Tap-changer position tens “3”	Operation indication signal common terminal	
CX - 15	Position display common terminal L	“1-N” operation indication signal	
CX - 16	Operation indication signal common terminal	“N-1” operation indication signal	
CX - 17	“1-N” operation indication signal	“Stop” operation indication signal	
CX - 18	“N-1” operation indication signal		
CX - 19	“Stop” operation indication signal		

6.4 CX-2 connection table

No.	Socket type	Terminal type	Remark
CX2 - 1	BCD code passive contact output "B01"	RS232 TX	Position signal output and voltage regulation command input of BCD code
CX2 - 2	BCD code passive contact output "B02"	RS232 RX	
CX2 - 3	BCD code passive contact output "B04"	RS232 GND	
CX2 - 4	BCD code passive contact output "B08"	RS485 A	
CX2 - 5	BCD code passive contact output "B10"	RS485 B	
CX2 - 6	BCD code passive contact output "B20"	Parallel port A	
CX2 - 7		Parallel port B	
CX2 - 8		BCD code passive contact output "B01"	
CX2 - 9		BCD code passive contact output "B02"	
CX2 - 10	BCD code passive contact output common terminal	BCD code passive contact output "B04"	
CX2 - 11	1-N command passive contact input	BCD code passive contact output "B08"	
CX2 - 12	Stop command passive contact input	BCD code passive contact output "B10"	
CX2 - 13	N-1 command passive contact input	BCD code passive contact output "B20"	
CX2 - 14	Voltage regulation command input common terminal	BCD code passive contact output common terminal	
CX2 - 15		1-N command passive contact input	
CX2 - 16		Stop command passive contact input	
CX2 - 17		N-1 command passive contact input	
CX2 - 18		Voltage regulation command input common terminal	
CX2 - 19		Analog output +	
CX2 - 20		Analog output -	

6.5 RS-485 connection table

DB-9	1	2	3	4	5	6	7	8	9
Signal	A	B							

6.6 Communication port connection table

DB-9	1	2	3	4	5	6	7	8	9
Signal	A	B							

Note: when two controllers are used in parallel, the A and B lines of the master and following units are connected respectively, and cross connection is not allowed.

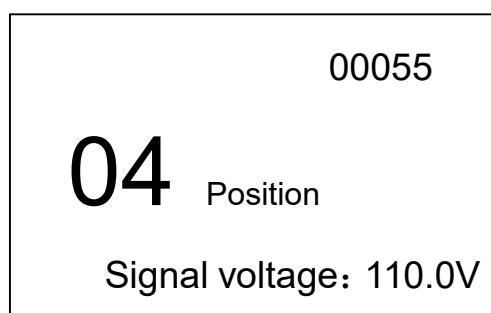
7. Use of operation keys

Caution! The operation key can only be used when the tap position display is correct. Before using the operation key, the control mode, delay time, pulse time, reference voltage and sensitivity must be set correctly (in the case of parallel control, the operation mode must also be set).

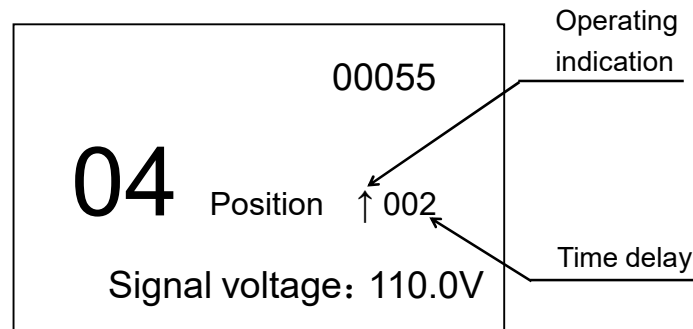
7.1 Manual operation

ET-SZ6 manual operation is realized by pushing the manual control keys on the panel (1-N, N-1 and stop buttons). Automatic control function is invalid during manual operation.

- Turn on the power switch at the back of ET-SZ6 and the home page will show on the LCD screen (as shown in the figure below)



- b) When it's manual control, the manual LED indicator is on. Press the "1-N" key, the "↑" indicator on the LCD screen is on, and the tap-changer goes up one position (as shown in the figure below)



- c) Press the "N-1" key, the "↓" indicator on the LCD screen is on, and the tap-changer goes down one position.
- d) In the process of raising or lowering positions, press the "Stop" key and the indicator flashes, the raising or lowering operation will stop immediately. The operation state in the above figure is indicated as "■".

7.2 Automatic operation

In the mode of automatic control, automatic LED indicator is on. ET-SZ6 will compare the signal voltage with the reference voltage. Once the signal voltage deviates from the set sensitivity and exceeds the set delay time, ET-SZ6 will send out a control signal of "1-N" or "N-1" (The indicator light for operation status will flash at the same time), making the tap-changer switch from one tap position to the next one, and realizing the purpose of voltage regulation.

When ET-SZ6 loses the tap position signal, the 1-N and N-1 operation commands will be automatically locked.

When the signal voltage is higher than the reference voltage upper limit, ET-SZ6 will send a quick drop operation command, and the overvoltage LED indicator will be on; When the signal voltage is lower than the reference voltage lower limit, ET-SZ6 will lock the operation and the overvoltage LED indicator will also be on.

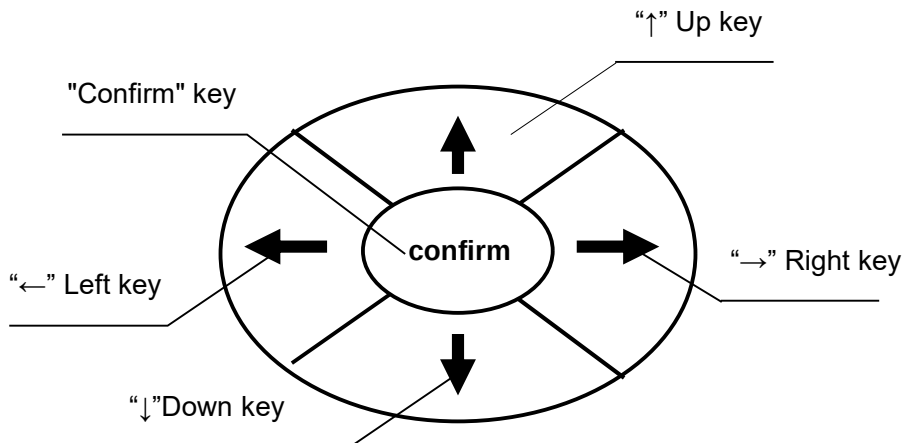
7.3 Remote control

Remote control includes passive contact input control and serial port control. The serial port control adopts RS-485 or RS-232 serial port bus communication mode, and the transmission protocol adopts MODBUS-RTU protocol (see the appendix for the communication protocol).

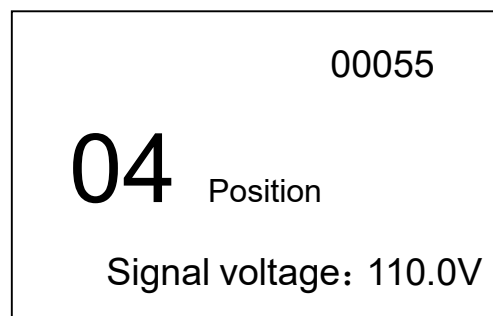
Note: in the manual control mode, both manual and remote control are valid.

8. Use of function keys

8.1 Name of function keys

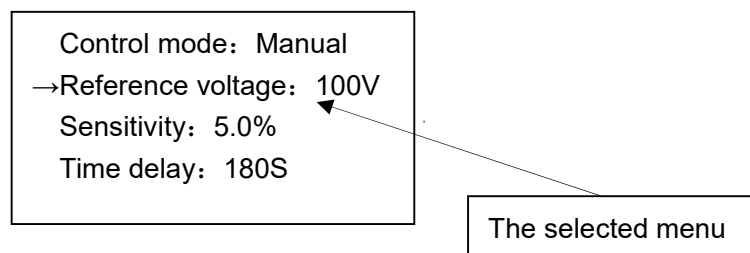


8.2 Home page on the screen

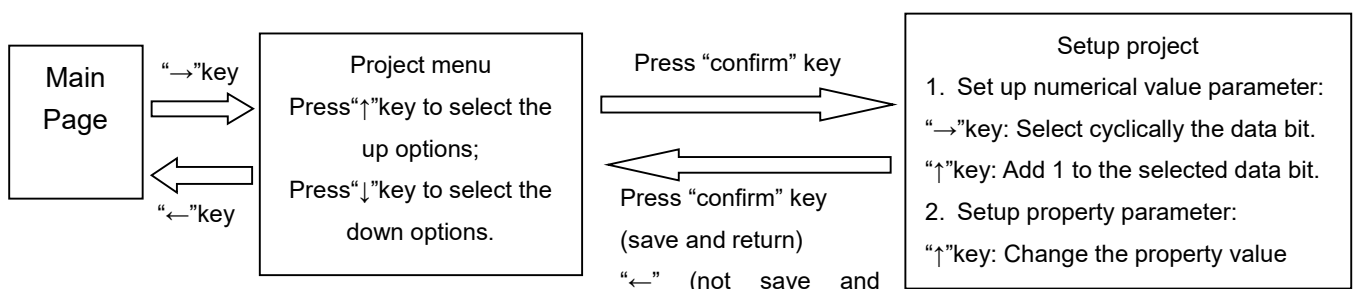


04—tap position signal; 00055—operation times; 110.0V—signal voltage

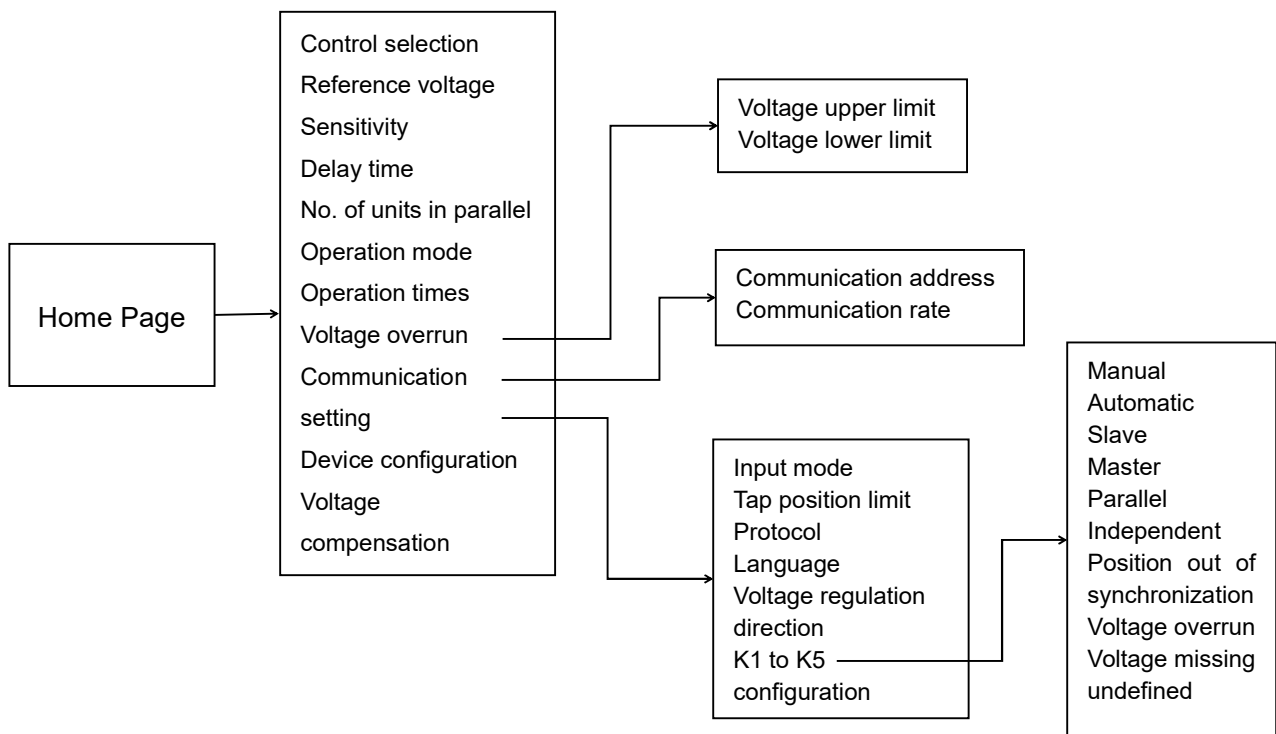
8.3 Setting menu on the screen



8.4 General operational sequence of the function keys

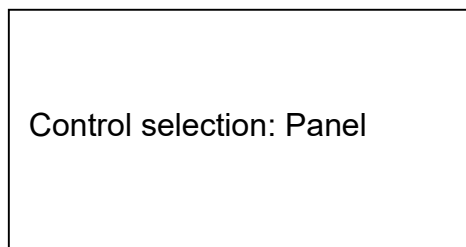


8.5 Structure of menu items



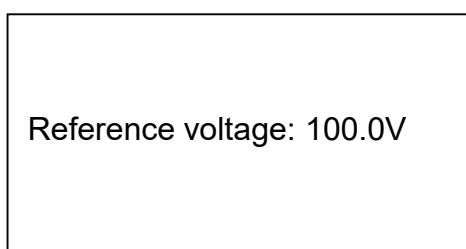
9. Menu explanation

9.1 Control selection



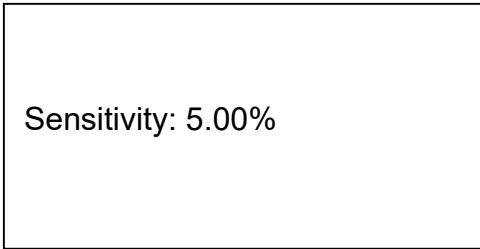
When "panel" is selected, "manual" or "automatic" is set by the front panel key. When "external" is selected, "manual" or "automatic" is input by the control mode and switched via dry contact.

9.2 Reference voltage



Under the automatic control mode, the signal voltage is compared with the reference voltage. When the deviation of the signal voltage exceeds the sensitivity setting range, the controller will adjust the position of the OLTC to make the signal voltage within the allowable range.

9.3 Sensitivity



Sensitivity: 5.00%

The sensitivity specifies the allowable range of signal voltage (i.e. voltage's upper limit and lower limit), with the following formula:

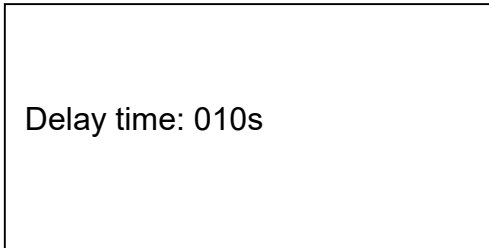
Upper limit = reference voltage + reference voltage × Sensitivity

Lower limit = reference voltage - reference voltage × Sensitivity

Example: if the user's reference voltage is 400V and the sensitivity is 2.5%, then the upper limit = $400V + 400V \times 2.5\% = 410V$

lower limit = $400V - 400V \times 2.5\% = 390V$

9.4 Delay time



Delay time: 010s

It is the out-of-range duration required for the controller to issue a regulation command when the signal voltage exceeds the automatic regulation setting range. The delay time is only valid in the automatic control mode and the setting time must be greater than the single operation time of the MDU.

9.5 No. of units in parallel

No. of units in parallel: 2

The total number of controllers in parallel operation, including master unit. It must be set correctly.

9.6 Operation mode

Operation mode: Independent

There are independent, master, slave 1, slave 2, slave 3, slave 4 and slave 5 six operation modes. In the transformer parallel operation application, there must be only one unit set as the master, and others are the slave. The slave must start from "slave 1". At this time, the slave units are controlled by the master unit.

9.7 Operation times

Operation times: 00006

It is used to set the operation times when using the controller for the first time.

9.8 Voltage overrun

Voltage upper limit: 120%
Voltage lower limit: 70%

It is used to set the upper and lower limit voltages which will trigger voltage overrun alarm. In automatic and manual mode, when input signal voltage is lower than 30V, the lower limit alarm function is invalid.

9.9 Communication setting

Communication address: 01H
Communication rate: 9600

- 1) Communication address: refer to the sole number used to identify the controller in RS-485 or RS-232 serial port communication network.
- 2) Communication rate: RS-485 or RS-232 serial port communication rate, including 1200 bit/s, 2400 bit/s, 4800 bit/s and 9600 bit/s.

9.10 Device configuration

Input mode: Decimal
Tap position upper limit: 35 positions
Protocol: Modbus
Language: English
Voltage regulation direction: 1-N

K1: voltage missing
K2: voltage overrun
K3: out of synchronization
K4: independent
K5: automatic

- 1) Input mode: Select tap position coding mode, decimal coding, BCD coding and line coding.
- 2) Tap position upper limit: Set the analog output signal 20mA's corresponding tap position (4mA corresponds to position 1).
- 3) Communication protocol: Select communication protocol: Modbus.
- 4) Language setting: select display language: Chinese or English.
- 5) Voltage regulation direction: select tap-changer operation direction in automatic mode: 1-N or N-1.
- 6) K1 to K5: Contact output configuration, It can be freely configured as voltage missing, voltage overrun, out of synchronization, independent, parallel, master, slave, automatic, manual, undervoltage, overvoltage.

9.11 Voltage compensation

Input Setting: off
Line Boost: 10%

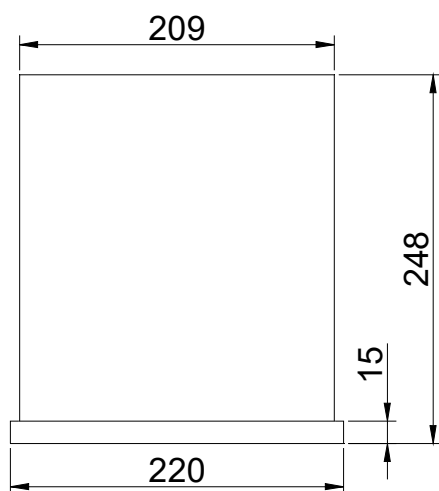
- (1) Input Setting: Set the voltage compensation function on or off.
- (2) Line Boost: Set the voltage compensation boost ratio from 1% to 29%.

9.12 Measuring parameter query

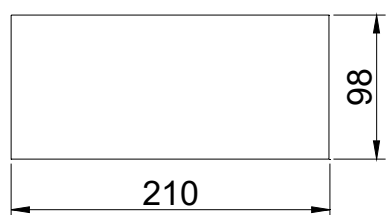
K1: ON voltage missing
 K2: ON voltage overrun
 K3: OFF out of synchronization
 K4: ON independent
 K5: OFF automatic

It displays the current working status and output signal definitions of output relays K1 to K5: ON or OFF.

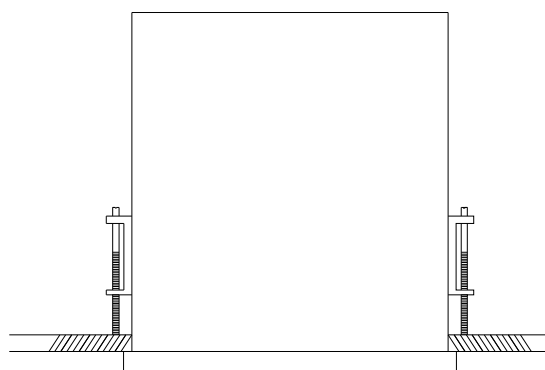
10. Installation method (mm)



Dimension



Installation



Installation

MODBUS Communication Protocol

1. MODBUS communication parameter

Item	Explanation
Baud rate	9600bps
Data type	Stop bit: 1bit Parity bit: None Data bits: 8bits
Data transmission sequence	From high to low

2. MODBUS protocol RTU frame format

MODBUS protocol is a master-slave communication method, where each communication is initiated by the master and the slave responds to the master's commands and send back data. ET-SZ6 adopts MODBUS RTU format (hexadecimal) and its frame structure is shown in Table 2.

Device address	Function code	Register address	Register length	CRC verification
8 bit	8 bit	16 bit	16 bit	16 bit

Table 1 Master RTU information frame

Device address	Function code	Response data	CRC verification
8Bit	8Bit	n x 8Bit	16Bit

Table 2 Slave RTU information frame

- (1) **Device address:** the communication address of ET-SZ6, there can't be two same addresses in one network.
- (2) **Function code:** Function code specified by MODBUS protocol. ET-SZ6 adopts function code 03 to read and save the register to realize data reading of tap position and operation times.
- (3) **Register address and register length:** The parameter in the master command is register from register address, and read N units of registers in the register length.
- (4) **Slave response data:** Number of bytes and N units of byte data

3. Modbus protocol command code definition

Modbus function code is defined as in Table 3.

Function code	Name
01	Read the coil's status
02	Read the input status
03	Read holding register
04	Read input register
05	Force single coil
06	Preset single register

4. Definition of registers

03 Read holding register

Register address	Modbus address	Data type	Description
00 00	40001	integer data	Tap position
00 01	40002	integer data	Number of operations
00 02	40003	integer data	Single voltage

For example

Master command: 01 03 00 00 00 03 05 CB

Slave return: 01 03 06 00 06 00 38 00 00 F6 0E

The master sends a command to read the data in the 3 registers starting from register 00 00.

The return of the slave is a hexadecimal number. 00 00 register corresponds to data 00 06, which represents that the tap position is 6. 00 01 register corresponds to data 00 38, which, converted to decimal, is 56 times. And 00 02 register corresponds to data 00 00, which represents that the current signal voltage is 0 V.

05 Force single coil

Coil address	Modbus address	Set command	Description
00 00	00001	FF 00	Operation command to raise tap position
00 01	00002	FF 00	Operation command to lower tap position

For example

Master command: 01 05 00 00 FF 00 8C 3A

Perform “set to 1” operation on coil address 00 00 to execute the operation command to raise the tap position.

Master command: 01 05 00 01 FF 00 DD FA

Perform “set to 1” operation on coil address 00 01 to execute the operation command to lower the tap position.

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